Information-optimal illumination for dead-time-limited single-pixel imaging:
certificates, water filling, and the jitter-capped ridge

[author block — user decision]

Abstract

Single-pixel imaging with photon-counting detectors is conventionally operated at low flux to keep dead-time loss small. A companion study mapped *when* deliberately raising the flux into the nonlinear dead-time regime helps a fixed family of illumination patterns, and located a finite-window count-information ridge at the cube-root load $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$ ($\nu = T/\tau$ dead-time slots per exposure) [1]. Here we ask two follow-up questions: what illumination extracts the most information from a dead-time-limited detector under fixed physical resources, and how hard a real, timing-jittered detector can be driven before that ridge collapses. We cast hand-designed sparse and matched patterns as atoms in a constrained experiment-design problem whose per-atom information is computed from the exact finite-window nonparalyzable renewal count law, and optimize a local D-optimal design under explicit detector-load, dose, per-pixel-exposure, and peak-irradiance constraints, with a generalized-equivalence / relaxed Kiefer–Wolfowitz certificate relative to a frozen structured dictionary. The design separates into two conjugate resource corners: a photon-budgeted variable-load optimizer (OED-DT) and a time-budgeted, power-available ridge-tracking policy (RIDGE-FIXED). On the theory side we derive an exact missing-information identity for hidden random hold times and the long-window count-information rate $J_\infty(\rho, c_v) = \rho/[(1+\rho)(1+c_v^2\rho^2)]$, whose optimum is finite for every nonzero hold-time variance and solves the cubic $c_v^2\rho^2(1+2\rho) = 1$, scaling as $2^{-1/3}c_v^{-2/3}$ at small jitter (c_v the hold-time coefficient of variation); matching this extensive loss to the deterministic terminal-phase loss predicts the smooth crossover $\rho^*(\nu, c_v) \sim (6\nu)^{1/3}(1+12\nu c_v^2)^{-1/3}$ — zero jitter is the critical manifold on which the ridge grows as $\nu^{1/3}$, and a Monte-Carlo sweep matches the exact-law peak predictions within 5–6% on the measured grid. We report a preregistered simulation campaign (M1) with a single gate on the optimized arm, a common charged pre-scan, six arms, and a mandatory per-cell disclosure ledger; all outcomes are reported as frozen [M1: primary/secondary verdicts and cross-arm numbers].

1 Introduction

Single-pixel imaging (SPI) and computational ghost imaging reconstruct a scene from the responses of a single bucket detector to a sequence of structured illumination patterns [2, 3, 4]. In the photon-starved regime the bucket is a photon-counting device that is blind for a dead time τ after each registered count, and the conventional response is to operate at low flux, keeping the per-slot load $\rho = \lambda\tau$ at the few-percent level and paying in acquisition time.

A companion study reframed high-flux operation as an operating-map question and answered it for fixed hand patterns [1]: dense multiplexed illumination is multiplex-limited, equal-load sparse illumination yields large but not uniformly broad gains, and the exact Fisher information of the integrated nonparalyzable count possesses a principal ridge at $\rho^*(\nu) = (6\nu)^{1/3} - 2/3 + O(\nu^{-1/3})$, beyond which no count-only estimator recovers information. That work took the illumination family as given. Two questions remain open. First, a *design* question: under a fixed incident-photon and dose budget, which illumination extracts the most information from the dead-time channel, and by how much can a certified optimizer improve on a strong fixed hand design? Second, a *deployment* question: the cube-root ridge is derived for an idealized detector with deterministic dead time; how far into the ridge can a real detector with timing jitter actually be driven?

We answer both by treating each hand-designed sparse or matched pattern as an atom in a constrained local experiment-design problem. Each atom's information is the exact chain-rule renewal information of Section 2, the design is optimized as a concave program over design measures, and the continuous optimum is certified against the frozen structured dictionary by a generalized-equivalence / relaxed Kiefer–Wolfowitz bound. The same formulation exposes a conjugacy: a photon-budgeted optimizer rations a fixed incident resource across directions (information water filling), whereas a time-budgeted policy that has power to spare tracks the ridge at every dwell. A parallel theory line asks how detector timing jitter modifies the ridge itself.

We formulate a local approximate-design framework for single-pixel illumination in which each pattern's information is computed from the exact finite-window nonparalyzable renewal count law, optimized under explicit detector-load, dose, per-pixel-exposure, and peak-irradiance constraints and accompanied by generalized-equivalence and exact-rounding certificates. Existing photon-counting optimal experiment design, adaptive or learned SPI, and dead-time optimal-flux studies address neighboring components [5, 6, 7, 8, 9, 10]; we found no prior framework combining these elements for integrated-count single-pixel illumination. The certificate is always stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model.

This work contributes: (i) a master concave design program over design measures with the exact renewal kernel, and a generalized-equivalence / relaxed Kiefer–Wolfowitz certificate against a frozen structured dictionary (Section 2); (ii) a photon-budget versus time-budget conjugacy that separates the design-allocation question (OED-DT) from the operating-point question (RIDGE-FIXED), with the associated information-water-filling reading; (iii) a random-hold count-information theory (Section 3): an exact missing-information identity for hidden iid hold times, the long-window rate $J_\infty(\rho, c_v) = \rho/[(1+\rho)(1+c_v^2\rho^2)]$ with its exact jitter-limited optimum $c_v^2\rho^2(1+2\rho) = 1$ and small-jitter law $2^{-1/3}c_v^{-2/3}$, and the matched crossover prediction $(6\nu)^{1/3}(1+12\nu c_v^2)^{-1/3}$, organized per the frozen unification ruling with the master design program as framework and the alignment result as its conditional proposition; and (iv) a preregistered simulation campaign (M1) whose scenes, atoms, load grid, guards, endpoints, and single gate were frozen in an immutable commit before any confirmatory reconstruction existed, reported as frozen including any negative primary (Sections 4–5) [M1: campaign outcomes]. All code, frozen preregistration documents, and per-cell evidence bundles are openly released.

2 The design problem

2.1 Master concave program over design measures

Freeze the coarse pre-scan reconstruction at $\hat{x}$ and parameterize the locally estimable image perturbation as $x(\theta) = \hat{x} + B\theta$, with $B \in \mathbb{R}^{n\times r}$ spanning a declared r-dimensional estimable subspace. For a nonnegative illumination atom a the arrival rate is $\lambda(a) = \Phi\, a^\top \hat{x} + d$ and the load is $\rho(a) = \tau\lambda(a)$. Let $J_{\mathrm{exact}}(\rho,\nu)$ be the exact per-slot Fisher information of the integrated nonparalyzable count for $\log\lambda$ derived in the companion work [1]. The score direction for θ is $q(a) = \Phi\, B^\top a/\lambda(a)$, so the correct rank-one local information atom is the chain-rule form

$$H(a;\hat{x}) = \nu\, J_{\mathrm{exact}}(\rho(a),\nu)\, q(a)q(a)^\top = \nu\, J_{\mathrm{exact}}(\rho(a),\nu)\, \frac{\Phi^2}{\lambda(a)^2}\, B^\top a a^\top B. \tag{1}$$

A raw sensitivity $g \propto \Phi a$ is incomplete unless the factor $\sqrt{\nu J_{\mathrm{exact}}}/\lambda$ is absorbed into it. For a probability measure ξ over the compact feasible atom set $\mathcal{A}$, define $V(\xi) = V_0 + \int_{\mathcal{A}} H(a)\, d\xi(a)$ with $V_0 \succeq 0$ the frozen pre-scan information, and maximize the local D-criterion

$$\max_{\xi}\ \log\det V(\xi) \quad \text{s.t. linear resource constraints,} \tag{2}$$

subject to average load, cumulative dose, family allocation, and peak constraints. The objective is concave in ξ; it is a *local information surrogate* and is not claimed to optimize reconstruction PSNR or any image-domain risk. The estimable subspace is frozen at $r = 200$: B is the top-200 eigenbasis of the fixed comparator's information matrix under the same pre-scan, dwell, and resource convention, computed before the optimization and held fixed, with a fixed numerical ridge $\epsilon_0 = 10^{-9}\,\mathrm{tr}(B^\top V_{\mathrm{FIXED}^*} B)/r$ (R13 §6). If every atom obeys a pointwise equal-load constraint the scalar kernel factors out and the geometry reduces to ordinary linear D-optimality [11]; genuine dead-time-aware design therefore requires controlled load variation.

2.2 The equivalence certificate

By concavity of $\log\det$ and affineness of $\xi \mapsto V(\xi)$, a feasible design ξ^* is optimal iff there exist constraint multipliers such that the reduced sensitivity $s(a) = \mathrm{tr}[V(\xi^*)^{-1}H(a)] - \sum_j \alpha_j e_j(a) - \sum_k \beta_k c_k(a)$ satisfies $s(a) \le \eta^*$ over $\mathcal{A}$ with equality on the support (the constrained Kiefer–Wolfowitz / general equivalence theorem for nonlinear models [11, 12]). Because a strict per-pixel-dose dual is not yet implemented, the frozen confirmatory quantity is the conservative additive budget-dual bound: with $d_a = \mathrm{tr}(V^{-1}M_{\mathrm{main}}H_a)$, $c_a = \rho_a$, budget $c^\top\xi \le b$, and $\theta \ge 0$,

$$G_{\mathrm{rel}} = \max_a (d_a - \theta c_a) - \sum_a \xi_a (d_a - \theta c_a) + \theta\{b - c^\top \xi\}, \qquad G_{\mathrm{rel}}/r \le 10^{-3}, \tag{3}$$

which upper-bounds the gap to the optimum of the relaxed (dose-free) problem and hence to the dose-constrained optimum (R13 §6.1). It is reported as `RELAXED_KW_UPPER_BOUND`, relative to the frozen dictionary and local estimate, not as a full per-pixel equivalence certificate. The maximum is taken by an exact scan of the frozen dictionary; a learned proposer may warm-start the search but the frozen run finishes with the exact oracle [13]. Continuous weights are rounded to an exact 972-row design by efficient rounding followed by deterministic exchange [14, 15, 16], admissible only if the exact D-efficiency ≥ 0.95.

2.3 Photon-budget versus time-budget conjugacy

The design admits two conjugate resource corners of the same dead-time information map. The gate-carrying OED-DT arm holds the original incident-signal resource as a hard *inequality*,

$$\int \rho_s(a)\, d\xi(a) \leq 0.60, \tag{4}$$

so a subset of high-value atoms may enter the ridge zone only if compensated by low-load or omitted atoms (R11 §4); detected counts are dual-reported but are never the sole fairness budget, because $\mathbb{E}[N \mid \rho] \simeq \nu\rho/(1+\rho)$ has vanishing marginal cost at high load. The no-gate RIDGE-FIXED arm instead fixes the support and applies one global multiplier per dwell, calibrated on the pre-scan estimate to mean load $\rho_R(\nu)$, followed by a single global safety clip; it uses a deliberately larger incident budget and is therefore descriptive (R11 §2). The exact production ridge is $\rho_R(\nu) = \min \arg\max_{\rho \in \mathcal{R}_{\mathrm{adm}}(\nu)} J_{\mathrm{exact}}(\rho, \nu)$ over the admissible set $\mathcal{R}_{\mathrm{adm}}(\nu) = \{0 < \rho \leq 24 : \; p_{\mathrm{ceil}} \leq 0.01,\; J_{\mathrm{mean}}/J_{\mathrm{exact}} \geq 0.90\}$, using the exact numerical ridge rather than the asymptotic cube-root form (R11 §1). Under the incident-photon budget the relevant scalar efficiency is $J_{\mathrm{exact}}(\rho, \nu)/\rho$, which decreases across the admitted palette; a photon-budgeted optimizer therefore rationally spends on lower loads and *avoids* the ridge, while the ridge belongs to the time-limited, power-available corner. This is a resource-regime statement, not a universal prohibition on ridge-zone allocation: sufficiently valuable directions may still receive larger loads in other cells (R13 §8(f)).

2.4 Guards as constraints

The raw optimum is admissible only after all frozen guards pass; each is a constraint of Eq. (2) or a post-rounding check, not a tunable proxy. *Peak (two-part, R13 A2):* a shape guard bounds unit-load support weights to $\frac{1}{4} \leq w_j \leq 4$ with support size $k \geq 16$, and a physical guard caps $\max_{i,j} a_{ij}^{\mathrm{physical}} \leq 1536$ (the peak of a uniform $k = 16$ atom at the absolute load cap $\rho = 24$), so matched-intensity ridge atoms cannot silently demand more peak power. *Weight power (R10 G4):* the dictionary contains powers $p \in \{0, 1\}$ only; $p = 2$ is excluded because dev evidence showed higher proxy contrast can worsen conditioning and endpoint quality. *Per-pixel dose (R10 G5):* each pixel's mean exposure stays within $\pm 5\%$ of the image mean, enforced in-optimizer and re-verified after rounding; a necessary row bound follows, $\rho_r \leq 1.05\, M\bar{d}/\max_j g_{rj}$ (R13 A5). *Spectral floor (R10 G6):* $V_{\mathrm{OED}} \succeq 0.5\, V_{\mathrm{FIXED}^*}$ on B, with a convex-mixture fallback whose coefficient is fixed from information matrices before any image endpoint is seen. *Kernel and trust (R11 G7–G8, R13 A1):* every (ν, ρ) node is certified for exact-PMF mass, score, and derivative agreement, mean-information efficiency $J_{\mathrm{mean}}/J_{\mathrm{exact}} \geq 0.90$, finite-window RQL log-load bias $|\log(\rho^\dagger/\rho)| \leq 0.05$, ceiling probability $p_{\mathrm{ceil}}(\rho, \nu) \leq 0.05$ per atom and ≤ 0.01 design-weighted. *A-risk (R10 G7):* $\mathrm{tr}(V_{\mathrm{exact}}^{-1}) \leq 1.05\, \mathrm{tr}(V_{\mathrm{FIXED}^*}^{-1})$. Any violation is a declared failure state (`DICTIONARY_RANK_FAIL`, `CONTINUOUS_CERTIFICATE_FAIL`, `SPECTRAL_GUARD_FAIL`, `A_RISK_GUARD_FAIL`, `DOSE_GUARD_FAIL`, `ROUNDING_FAIL`), whose predeclared fallback is the fixed comparator; no threshold is retuned after freeze.

3 Theory: the jitter-capped ridge and its unification

The theory spine follows the frozen unification ruling: the random-hold count-information results of Section 3.1 are the headline theorems, the master design program of Section 3.2 is the unifying framework, and the brightness–information alignment of Section 3.3 is the conditional proposition explaining RIDGE-FIXED. The deterministic detector of the companion [1] is the zero-jitter critical manifold of the random-hold theory. In one sentence: the recovery-time law determines a scalar count-information kernel; constrained design allocates illumination directions and loads against that kernel; and fixed global flux is the special case in which scene brightness already satisfies the kernel's water-filling condition. Theorem statements are given here; proofs at manuscript rigor are deferred to the supplement [supplement: proofs of Theorems 1–3 and Proposition 1].

3.1 The jitter-capped ridge

Generalize the detector so that after each registered event it blocks for a hidden *random* hold time $B_i \sim H$, iid, independent of the Poisson arrivals and of λ, with mean τ and coefficient of variation $c_v = \sigma_B/\tau$ (the companion's deterministic model is $c_v = 0$). The observed datum remains the integrated registered count N_T over $T = \nu\tau$; the active time L_T, the state path, and the realized holds are hidden from the count-only observer.

Theorem 1 (missing-information identity) *For $\theta = \log \lambda$, the complete detector-path information is*

$\mathbb{E}[N_T]$*, and the information retained by the scalar count is exactly*

$$I_N(\theta;T) = \mathbb{E}[N_T] - \lambda^2\,\mathbb{E}\big[\mathrm{Var}(L_T \mid N_T)\big] \qquad (5)$$

for every finite T *and every iid hidden hold law* H *independent of* λ.

The companion's terminal-phase identity is the special case in which the uncertainty in L_T is confined to the terminal detector phase. The identity requires amendment if the hold law depends on λ or the scene, if realized holds are observed and used, if holds are serially correlated, or if the counter is paralyzable.

Theorem 2 (long-window rate) *Under standard nonlattice renewal local-limit conditions,*

$$\frac{I_N(\log\lambda;T)}{\nu} \longrightarrow J_\infty(\rho,c_v) = \frac{\rho}{(1+\rho)\,(1+c_v^2\rho^2)}, \qquad (6)$$

depending on H *only through its mean and variance. The missing-information rate converges to* $c_v^2\rho^3/[(1+\rho)(1+c_v^2\rho^2)]$*: an* extensive *loss equal, to first order in* $c_v^2\rho^2$*, to the heuristic* $\lambda^2\sigma_B^2\,\mathbb{E}[N_T]$ *— conditioning on the count partially reveals the accumulated hold fluctuation, contributing the factor* $(1+c_v^2\rho^2)^{-1}$ *— and, unlike the deterministic terminal-phase loss, nonvanishing as* $\nu\to\infty$.

Theorem 3 (exact jitter-limited optimum) *For every fixed* $c_v > 0$*,* $J_\infty(\rho,c_v)$ *has a unique interior maximum* ρ_c *solving the cubic*

$$c_v^2\,\rho_c^2\,(1+2\rho_c) = 1, \qquad (7)$$

with small-jitter expansion

$$\rho_c = 2^{-1/3}c_v^{-2/3} - \tfrac{1}{6} + \tfrac{2^{1/3}}{36}\,c_v^{2/3} + O(c_v^{4/3}), \qquad (8)$$

whose leading coefficient is $2^{-1/3} = 0.79370\ldots$

For exponential holds ($c_v = 1$, outside the small-jitter regime) the exact positive root is $\rho_c = 0.6573\ldots$; gamma, bounded two-point, and lognormal holds sharing a small c_v share the leading coefficient, with distribution shape entering at the next order.

Matched crossover (prediction). Matching the extensive loss $c_v^2\rho^2$ against the deterministic terminal-boundary loss $\rho^2/(12\nu)$ in the joint regime $\nu\to\infty$, $c_v\to 0$, $c_v^2\nu = O(1)$ gives

$$\rho_*(\nu,c_v) \sim \left(2c_v^2 + \frac{1}{6\nu}\right)^{-1/3} = (6\nu)^{1/3}\,\big(1+12\nu c_v^2\big)^{-1/3}, \qquad (9)$$

with scaling variable $x = 12\nu c_v^2$ and crossover at $c_v \asymp (12\nu)^{-1/2}$: the deterministic limit $x\to 0$ recovers $(6\nu)^{1/3}$ and the jitter-dominated limit recovers $2^{-1/3}c_v^{-2/3}$. An earlier $\min\{(6\nu)^{1/3},\, c_0\,c_v^{-2/3}\}$ conjecture is only the two-limit envelope of Eq. (9), with an artificial kink, and is superseded. Per the frozen claim discipline, the exact identity and the long-window limit *imply* the jitter exponent $-2/3$; the full finite-window crossover of Eq. (9) is a matched-asymptotic *prediction* until a uniform two-parameter (triangular-array renewal local-limit) proof is complete [supplement: uniform crossover proof, or the prediction label is retained].

Table 1: Monte-Carlo scalar count information $J(\rho,\nu=2000,c_v)$ for lognormal hold times (coarse sweep: common random numbers, 60,000 frames), validated against the exact table at $c_v = 0$ ($J(2) = 0.676$ vs. exact 0.667 + corr), against the exact-law peak predictions of Theorem 3/Eq. (9) as registered before the refined sweep completed (`docs/R14_PREREGISTERED_PREDICTIONS.md`, commit `8a627bb`). The coarse-grid fit $c_0 \approx 0.72$ is an effective finite-grid, finite-sample coefficient; the exact asymptotic coefficient is $2^{-1/3} = 0.794$, and the exact-law predictions sit 5–6% above the coarse measurements (grid-limited; the ratio test $\mathrm{cap}(0.05)/\mathrm{cap}(0.1) = 1.65$ vs. $2^{2/3} = 1.587$ is less sensitive to the common bias). Hardware warning: at $\rho = 24$ with $c_v = 0.05$, $J = 0.38$ vs. deterministic 0.95 — about 60% information loss from 5% jitter at the deterministic ridge. Source: `results/round63_study2/jitter_scalar_fi.log`.

c_v	measured peak ρ	exact-law prediction	J at peak
0	≈ 22–24	22.25 (exact ridge)	0.948
0.05	≈ 5.43	5.69	0.796
0.1	≈ 3.30	3.50	0.701
0.3	≤ 2 (grid edge)	1.63	0.499

Universality class. The exponent pair $(1/3, -2/3)$ is universal within the regular finite-variance iid hidden-hold class: iid hidden nonnegative holds whose law is independent of λ, with finite nonzero mean and finite variance, forming a regular small-jitter family, under count-only observation and active-start nonparalyzable operation. It is *not* claimed for heavy-tailed (infinite-variance) holds, correlated or drifting holds, holds observed by the electronics, paralyzable detectors, event-time observations, or λ-dependent recovery. The precise critical statement is: zero hold-time variance is the critical manifold on which the optimal load continues to grow with window length; any fixed nonzero hidden hold variance produces a finite long-window optimum. Adjacent renewal-CLT and random-dead-time literature establishes the mean/variance ingredients and task-specific finite-rate optima [17, 18, 19, 20, 10]; we found no source stating the count-only exponent pair or the extensive/terminal crossover.

The coarse sweep of Table 1 was measured *be-*

fore the exact law was derived; the exact-law column was then registered, together with two out-of-sample rows — $c_v = 0.02 \to \rho_* \approx 10.4$ and $c_v = 0.2 \to \rho_* \approx 2.30$ — before the refined sweep completed (`docs/R14_PREREGISTERED_PREDICTIONS.md`, commit `8a627bb`). The refined sweep (three hold families — lognormal, matched-CV gamma, bounded two-point — with cross-fitted conditional-score information estimation, frozen exponent tests for $1/3$ and $-2/3$, a long-window cubic residual, and a scaling-collapse test of Eq. (9)) reports in Section 5.

3.2 One master concave program

The framework theorem generalizes Section 2: with the random-hold kernel $J_H(\rho, \nu)$ of Theorem 2 inside the rank-one atoms $H_z = \nu J_H(\rho_z, \nu)\, q_z q_z^\top$, the design-measure problem of Eq. (2) remains concave, and every true interior optimum of the declared problem satisfies one generalized reduced-sensitivity/KKT system: the sensitivity $\psi(z) = M\,\mathrm{tr}[V^{-1}H_z] - \sum_m \beta_m c_m(z)$ obeys $\psi(z) \le \eta$ with equality on support atoms, and at any support atom interior in a smooth scalar coordinate y, $\partial_y \psi = 0$ with $\partial_y^2 \psi \le 0$. The load coordinate yields the information-water-filling equation — $M\nu J_H'(\rho)\,\ell$ equal to the marginal resource shadow price, with $\ell = q^\top V^{-1} q$ — while the support-scale and weight-power coordinates equate $2M\nu J_H(\rho)\, q_y^\top V^{-1} q$ to the corresponding marginal dose/peak/dispersion price: the apparent "conditioning penalty" is not an ad hoc term but the V^{-1} geometry inside the reduced sensitivity, which also explains why raising proxy contrast from $p = 1$ to $p = 2$ can lower the true objective. One caveat is kept deliberately: the master design problem places load, support scale, and intensity weighting in one constrained concave measure optimization, and its support atoms satisfy a common reduced-sensitivity condition; but the empirically observed occupancy ($k \approx 32$) and weighting ($p \approx 1$) optima *motivate* these coordinates and are not themselves claimed as continuous KKT theorems — they were selected from discrete grids and judged by image endpoints, not by the master log det objective.

3.3 Brightness–information alignment

At a fixed design iterate with directional values $\ell_i = q_i^\top V^{-1} q_i$, allocate scalar loads under the budget $\sum_i \pi_i \rho_i = B$ to maximize $F(\rho) = \sum_i \pi_i \ell_i J(\rho_i)$; on an interval where J is increasing and strictly concave, the water-filling optimum satisfies $\ell_i J'(\rho_i^*) = \beta$ at every interior coordinate. A single global source induces $\tilde\rho_i = \kappa u_i$, with $u_i = a_i^\top x$ the bucket brightness.

Proposition 1 (alignment and residual bound) *The fixed-flux allocation is exactly optimal for this scalar subproblem if and only if there exists β with*

$$\ell_i\, J'(\kappa u_i) = \beta \tag{10}$$

for all interior active directions, with KKT-compatible inequalities at the bounds. If moreover $-\ell_i J''(\rho) \ge m > 0$ throughout the admitted interval, then, with residuals $r_i = \ell_i J'(\kappa u_i) - \beta$,

$$0 \;\le\; F(\rho^*) - F(\tilde\rho) \;\le\; \frac{1}{2m} \sum_i \pi_i\, r_i^2. \tag{11}$$

A computable projected-KKT residual — not C_u alone — therefore certifies whether one global flux knob is close to the scalar water-filling optimum. In power-law form, if $J'(\rho) \asymp C\rho^{-s}$ and $\ell_i \asymp u_i^\alpha$, fixed flux aligns when $\alpha \approx s$; for the rising kernel $J = \rho/(1+\rho)$ the local slope $s(\rho) = 2\rho/(1+\rho)$ runs from 0 toward 2, so the often-invoked $\ell \propto u^2$ alignment is only a high-load approximation. Without the alignment condition there is no scene-independent constant-factor guarantee: two directions with $u_1/u_2 \to \infty$ but $\ell_1/\ell_2 \to 0$ drive the fixed-flux objective ratio to zero, so fixed flux can be arbitrarily bad. With random holds, J' vanishes at the jitter cap and is negative beyond it, so fixed flux can overdrive the brightest directions and explicit gain control becomes necessary. This explains both sides of the companion's dev evidence [1]: RIDGE-FIXED with one global knob is near-optimal exactly when its realized residuals are small (self-water-filling), while the uniform servo — which loses by up to ~ 7 dB on sparse-support scenes — fails because forcing equal ρ_i discards a naturally favorable alignment between u_i and ℓ_i.

4 Preregistered campaign (M1)

M1 is a separate follow-up campaign with its own seed namespace, manifests, and expected-cell ledger; it does not reopen the companion's frozen studies. It freezes on the R13 audit plus one immutable commit (tag `m1-freeze`); after that commit no scene, atom family, load grid, guard, endpoint, or gate may change, and if the primary fails there is no third redesign.

Geometry and constants. $n = 32^2 = 1024$; $M_{\text{total}} = 1024 = M_{\text{pre}}(52) + M_{\text{main}}(972)$ for every arm; nonparalyzable active start, $\tau = 50$ ns, $d = 0$, $\sigma_b = 0$; $\bar\rho \in \{0.05, 0.60\}$ for the budget-anchored arms; $\nu \in \{5, 10, 20, 50, 100, 200, 500, 1000, 2000\}$; five measurement seeds per (image, condition). Optical time $T_{\text{opt}} = M_{\text{total}}\, \nu\, \tau$; design-computation wall time is reported separately and never folded into T_{opt}.

Scenes. Twenty-four fresh confirmatory instances (six generator families $\times$ four instances) with seeds $s_{f,r} = 633000 + 100f + r$, plus six development-only instances $632900 + f$. The six committed generators (glyph, chirp, spokes, maze, contour, microtexture) are reused verbatim with their actual names — "USAF" is not used unless a USAF target is generated, and spokes is described as a radial resolution target. No acceptance filter of any kind is applied; generators, images, and ROIs receive SHA256 manifests.

Common pre-scan. Every arm receives and pays for the same 52-pattern scene-independent balanced pre-scan at its own $(\bar{\rho}, \nu)$: 32 paired 4×4-block rows (support 32, amplitude 32), 16 block rows (amplitude 16), and 4 quadrant rows (amplitude 4), so every pixel receives cumulative amplitude $32 + 16 + 4 = 52$ and the row-mean pattern is exactly the all-ones vector; the mean pre-scan load then equals the declared global load for any sum-normalized scene, with no $\hat{x}$-dependent row scaling (R13 §2). All 52 counts enter every arm's final reconstruction; only adaptive arms use the pre-scan to choose subsequent patterns. Fixed arms run 972 main patterns; every arm totals exactly 1024 physical exposures.

Arms. Table 2 lists the six arms. Only OED-DT carries a confirmatory gate; the others are ablations, baselines, or the descriptive ridge-tracking demonstration. The R13 freeze audit additionally mandates MATCH1 as a no-gate arm and a committed selection of the global fixed comparator FIXED* from {SCAT32, LBLOB16, MATCH1} by the frozen development rule (median fixed-dwell radiometric PSNR at $\bar{\rho} = 0.60$, $\nu = 2000$); LBLOB16 is the current spectral-floor reference [M1: committed FIXED* selection record]. Reconstruction (RQL with the frozen analytic λ_{TV} rule and pooled initialization) is identical across arms; the primary comparison is each arm's own fast curve against the common safe ($\bar{\rho} = 0.05$) reference of its own pattern set.

Endpoints and gate. The frozen Q_{90} time-to-quality machinery transfers verbatim: seed-mean radiometric PSNR, equal-weight PAVA isotonic curves, a 0.50 dB safe-range informativeness floor, the frozen censoring taxonomy, and a 10,000-replicate nested family-stratified bootstrap. The single primary gate applies to OED-DT only (`METHOD_SPEED_PASS`): median $S_{\mathrm{gate}} \geq 3$, bootstrap lower bound > 1, and $\geq 18/24$ scenes with $S_j > 1$. Two secondaries cannot rescue a failed primary: a fixed-dwell safe-versus-fast gain (`METHOD_FIXED_DWELL_PASS`; median ≥ 1 dB at the terminal dwell, lower bound > 0, $\geq 18/24$ positive) and a design gain of OED-DT over FIXED* at the fast terminal dwell (`METHOD_DESIGN_PASS`; median ≥ 0.50 dB, family-stratified lower bound > 0, $\geq 18/24$ positive).

Table 2: The six M1 arms. Only OED-DT carries the confirmatory gate; the R13 freeze audit additionally requires MATCH1 and a committed FIXED* selection (see text).

Arm	Role	Gate
OED-DT	variable-load exact-kernel D-optimal design (v3 solver; relaxed-KW certificate vs. the frozen dictionary)	primary
OED-EQLOAD	same optimizer, all atoms at $\bar{\rho}$ (kernel ablation)	none
SCAT16	frozen scattered $k = 16$ (companion bridge)	none
SCAT32	best ladder rung by fixed-dwell median	none
LBLOB16	committed clustered support (strongest fixed baseline; spectral-floor reference)	none
RIDGE-FIXED	dwell-dependent ridge-tracking policy (time-budget demonstration)	none

Disclosure ledger. Every cell discloses incident and detected photons (predicted and realized), dose ratios, source/peak-irradiance ratio, the load percentiles ρ_5, ρ_{50}, ρ_{95}, $\rho_{\max}$, predicted and realized ceiling fractions, and mean J_{exact}; designed cells add the continuous and exact $\log\det$ objectives, the exact D-efficiency, A-risk, spectral floor, and the relaxed-KW gap. The full nine-dwell kernel/ridge certification table is frozen before any confirmatory scene is opened.

5 Results

5.1 Primary: OED-DT time-to-quality

[M1: METHOD_SPEED_PASS verdict — median S_{gate}, bootstrap lower bound, count of scenes with $S_j > 1$, and the frozen PASS/FAIL call.]

5.2 Design gain: OED-DT vs. FIXED*

[M1: METHOD_DESIGN_PASS — median design gain (dB) at the fast terminal dwell, family-stratified lower bound, count positive.]

5.3 Fixed-dwell safe-versus-fast gain

[M1: METHOD_FIXED_DWELL_PASS — median terminal-dwell gain (dB), count positive, bootstrap lower bound, per-arm ladder.]

5.4 RIDGE-FIXED operating-point demonstration

[M1: fixed-dwell gain per dwell of the ridge-tracking policy over $\bar{\rho} = 0.60$ and safe references, realized loads, and ceiling fractions.]

5.5 OED-EQLOAD kernel ablation

[M1: equal-load geometry vs. variable-load geometry — any design difference attributable to the renewal kernel, with numerical-noise disclosure.]

5.6 Certificate and rounding diagnostics

[M1: relaxed-KW gap G_{rel}/r, exact D-efficiency, A-risk ratio, and spectral-floor margin per designed cell.]

5.7 Jitter-ridge refined sweep

[sweep: refined-sweep verdict — deterministic 1/3 and jitter $-2/3$ exponent tests, long-window cubic residual, scaling collapse of $\rho_*/(6\nu)^{1/3}$ against $x = 12\nu c_v^2$ per Eq. (9), distribution-shape universality across lognormal/gamma/two-point holds, and the registered out-of-sample rows $c_v = 0.02 \to \rho_* \approx 10.4$ and $c_v = 0.2 \to \rho_* \approx 2.30$.]

6 Discussion

The two arms expose conjugate resource regimes of the same dead-time information map. Under a linear incident-photon budget the per-photon efficiency $J_{\text{exact}}(\rho, \nu)/\rho$ falls monotonically, so a budget-constrained design rations photons onto lower loads and *avoids* the ridge; the ridge belongs to time-constrained operation, where power is available and the only scarcity is dwell. Reading the flux ridge and the design allocation as one program clarifies that the large fixed-dwell gains of the companion are an operating-point effect, not evidence that a variable-load optimizer must spend into the ridge.

The jitter-capped ridge is the practical caution. The cube-root ridge is a property of the zero-jitter critical manifold; the moment a detector's hold time carries a coefficient of variation c_v, the long-window optimum is finite — the exact cubic $c_v^2\rho^2(1+2\rho) = 1$ of Theorem 3, scaling as $2^{-1/3}c_v^{-2/3}$ — and driving past it is actively harmful: at the deterministic ridge, 5% jitter already forfeits about 60% of the scalar count information in the coarse dev measurement (Table 1), with the final uncertainty of that loss metric assigned by the preregistered batch protocol of the refined sweep. A deployment must therefore locate its operating point against the jitter-limited ceiling of the real detector, not the deterministic ridge; the safe conclusion is conditional on measured timing jitter.

An optical extension follows the companion's discussion: a coarse complex-field estimate could synthesize an optical displacement that suppresses the predicted common mode before photon counting, removing photons ahead of dead-time loss rather than after it [1]. All results here are simulation-based, and every certificate is stated relative to the frozen dictionary, the local pre-scan estimate, and the declared resource model; optical-bench validation remains outstanding.

7 Reproducibility

The design formulation, guards, and campaign follow four frozen rulings: the dead-time OED formulation and equivalence certificate (R10), the ridge-zone load architecture and incident-budget conjugacy (R11), the M1 freeze audit (R13), and the unification ruling (R14, issue "R14 ruling: unification architecture"), whose theorem hierarchy Section 3 follows. Exact-law peak predictions for the refined jitter sweep, including the two out-of-sample rows, were registered before the sweep completed (`docs/R14_PREREGISTERED_PREDICTIONS.md`, commit `8a627bb`). The campaign freezes at tag `m1-freeze` once the R13 checklist passes outcome-blind self-tests; the companion is frozen at tag `paper1-freeze-v1`. Source-of-record documents for every frozen constant are cited by ruling number in the block comments of the manuscript source. All code, frozen preregistration documents, per-cell manifests, and evidence bundles are available at [repo URL wording — user decision]; funding and acknowledgments are [funding/acknowledgments — user decision].

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Table 3: Cross-arm summary (descriptive; no new gates beyond the OED-DT primary). All entries fill from the frozen M1 analyzer after tag `m1-freeze`.

Arm	median S_{gate}	design (dB)	fixed-dwell (dB)	mean J_{exact}	incident budget	ceiling frac.	verdict
OED-DT	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
OED-EQLOAD	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
SCAT16	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
SCAT32	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
LBLOB16	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]
RIDGE-FIXED	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]	[M1]

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